Denoting the array factor by

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jikd\cos\theta} \right|^2, \tag{9.101}$$

the power density of the antenna array is then written as

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi)\, F_a(\theta). \tag{9.102}$$

This equation demonstrates the ***pattern multiplication principle***. It allows us to find the far-field power density of the antenna array by first computing the far-field power pattern with the array elements replaced with isotropic radiators, which yields the array factor $F_a(\theta)$, and then multiplying the result by $S_e(R_0, \theta, \phi)$, which is the power density for a single element (and is the same for all elements).

The feeding coefficient A_i is, in general, a complex amplitude consisting of an amplitude factor a_i and a phase factor ψ_i:

$$A_i = a_i e^{j\psi_i}. \tag{9.103}$$

Insertion of Eq. (9.103) into Eq. (9.101) leads to

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jikd\cos\theta} \right|^2. \tag{9.104}$$

The array factor is governed by two input functions: the ***array amplitude distribution*** given by the a_i's and the ***array phase distribution*** given by the ψ_i's.

▶ The amplitude distribution serves to control the shape of the array radiation pattern, while the phase distribution can be used to steer its direction. ◀

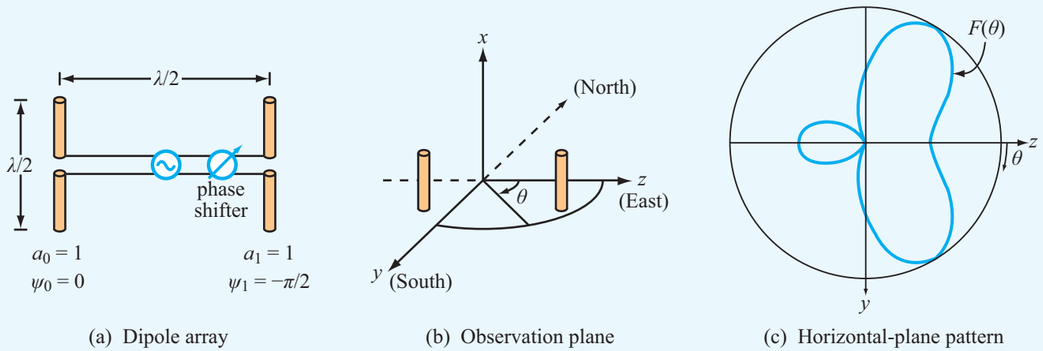
Example 9-5: Array of Two Vertical Dipoles

An AM radio station uses two vertically oriented half-wave dipoles separated by a distance of $\lambda/2$, as shown in **Fig. 9-28(a)**. The vector from the location of the first dipole to the location of the second dipole points toward the east. The two dipoles are fed with equal-amplitude excitations, and the dipole farther east is excited with a phase shift of $-\pi/2$ relative to the other one. Find and plot the antenna pattern of the antenna array in the horizontal plane.

Solution: The array factor given by Eq. (9.104) was derived for radiators arranged along the z axis. To keep the coordinate system the same, we choose the easterly direction to be the z axis, as shown in **Fig. 9-28(b)**, and we place the first dipole at $z = -\lambda/4$ and the second at $z = \lambda/4$. A dipole radiates uniformly in the plane perpendicular to its axis, which in this case is the horizontal plane. Hence, $S_e = S_0$ for all angles θ in **Fig. 9-28(b)**, where S_0 is the maximum value of the power density radiated by each dipole individually. Consequently, the power density radiated by the two-dipole array is

$$S(R, \theta) = S_0 F_a(\theta).$$

For two elements separated by $d = \lambda/2$ and excited with equal amplitudes ($a_0 = a_1 = 1$) and with phase angles $\psi_0 = 0$ and



(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

Figure 9-28 Two half-wave dipole array of Example 9-5.

将数组因子表示为

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jikd\cos\theta} \right|^2, \tag{9.101}$$

天线阵列的功率密度可写为

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi)\, F_a(\theta). \tag{9.102}$$

该方程演示了模式乘法原理。它允许我们首先计算用各向同性辐射器替换阵列元件的远场功率方向图，从而得出阵列因子 $F_a(\theta)$，然后将结果乘以 $S_e(R_0、\theta、\phi)$，这就是单个元件的功率密度（并且对于所有元件都相同），从而找到天线阵列的远场功率密度。

一般来说，馈电系数 A_i 是一个复振幅，由振幅因子 a_i 和相位因子 ψ_i 组成：

$$A_i = a_i e^{j\psi_i}. \tag{9.103}$$

插入方程。 (9.103) 代入方程。 (9.101) 导致

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jikd\cos\theta} \right|^2. \tag{9.104}$$

阵列因子由两个输入函数控制：由 a_i 给出的阵列幅度分布和由 ψ_i 给出的阵列相位分布。

▶ 幅度分布用于控制阵列辐射方向图的形状，而相位分布可用于控制其方向。 ◀

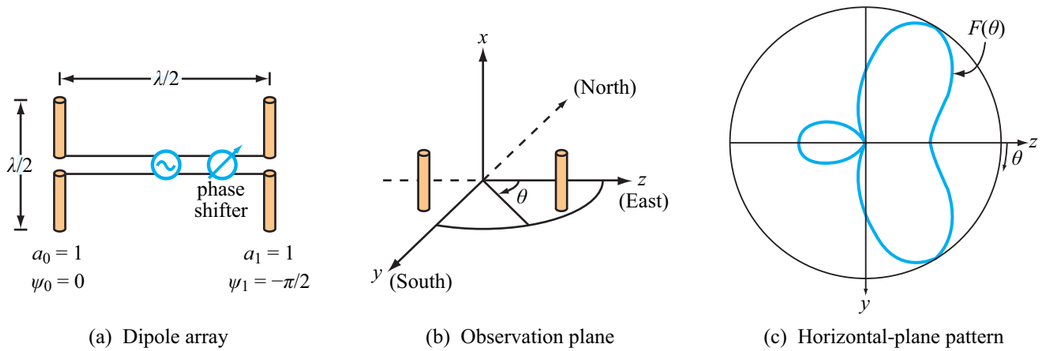
例 9-5： 两个垂直偶极子的阵列

AM广播电台使用两个垂直定向的半波偶极子，其间隔距离为$\lambda/2$，如图9-28(a)所示。从第一个偶极子位置到第二个偶极子位置的矢量指向东方。两个偶极子受到等幅激励，更东的偶极子受到相对于另一个偶极子 $-\pi/2$ 相移的激励。查找并绘制水平面上天线阵列的天线方向图。

解：由方程给出的阵列因子。 (9.104) 是针对沿 z 轴排列的散热器得出的。为了保持坐标系相同，我们选择东向作为z轴，如图9-28(b)所示，并将第一个偶极子放置在z $= -\lambda/4$处，第二个偶极子放置在z$= \lambda/4$处。偶极子在垂直于其轴的平面（在本例中为水平面）内均匀辐射。因此，图9-28(b)中所有角度θ的$S_e = S_0$，其中S_0是每个偶极子单独辐射的功率密度的最大值。因此，双偶极子阵列辐射的功率密度为

$$S(R, \theta) = S_0 F_a(\theta).$$

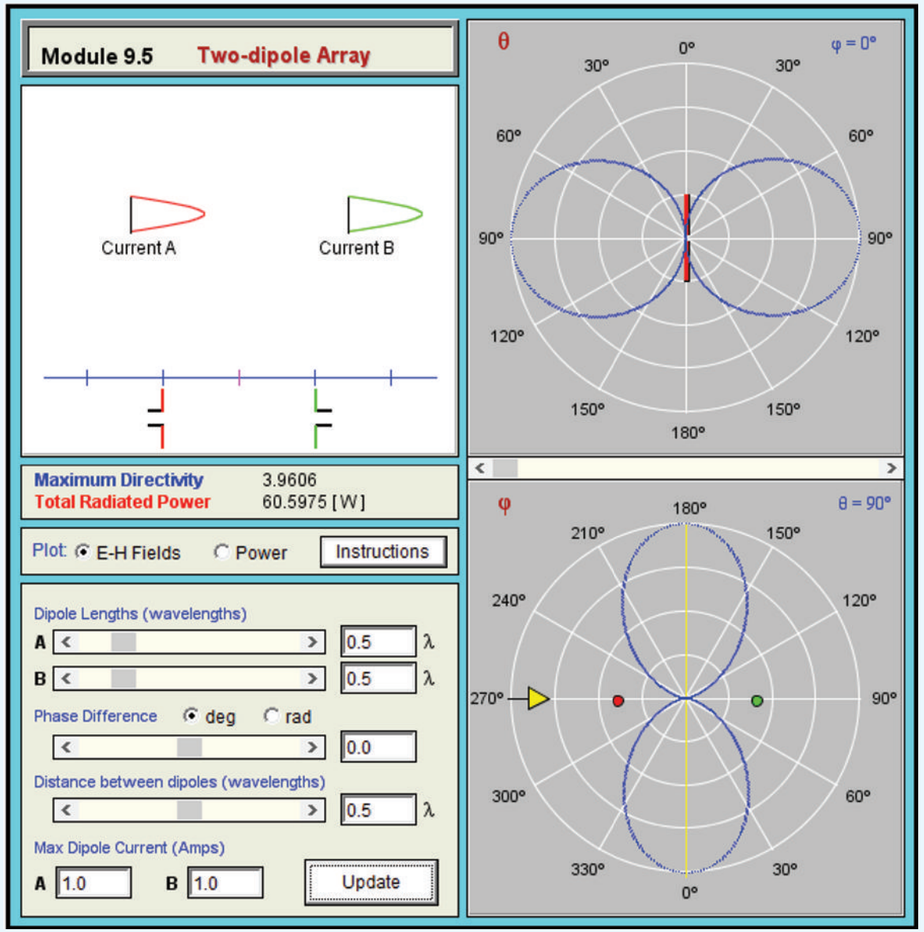
对于由 $d = \lambda/2$ 分隔并以相等幅度 ($a_0 = a_1 = 1$) 和相角 $\psi_0 = 0$ 激励的两个元素，



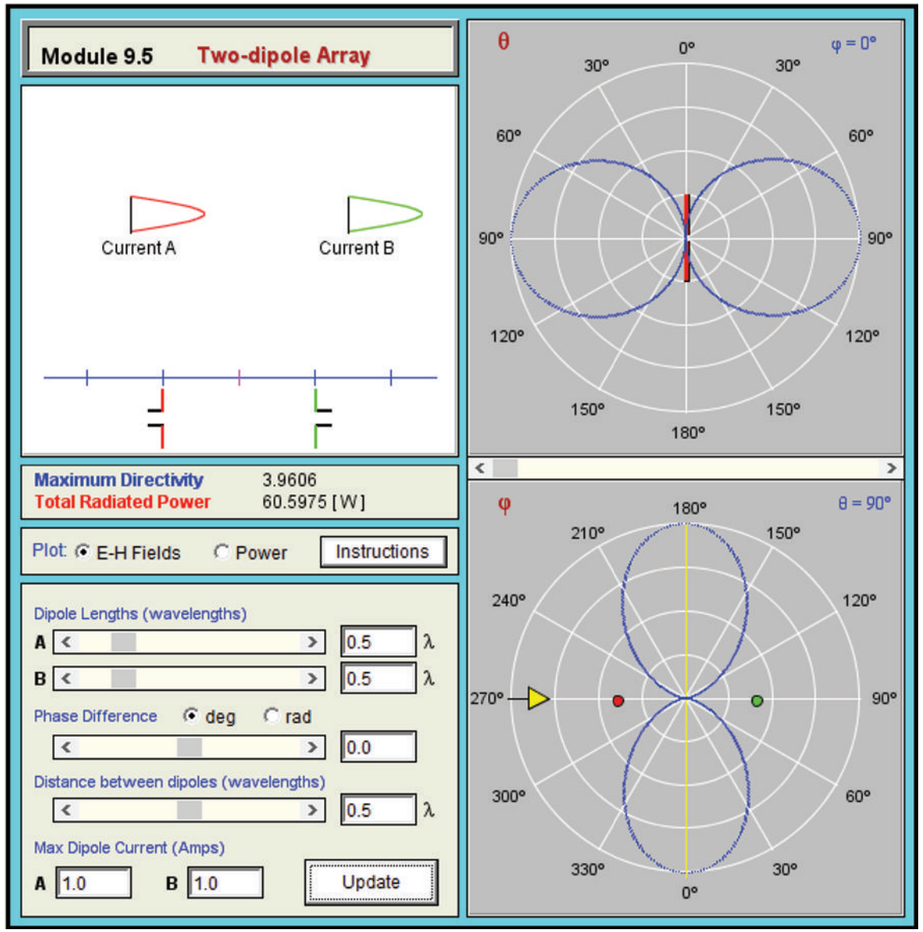
(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

图9-28 例9-5的两个半波偶极子阵列。

Module 9.5 Two-Dipole Array Given two vertical dipoles, the user can specify their individual lengths and current maxima, as well as the distance between them and the phase difference between their current excitations. The module generates plots of the field and power patterns in the far zone and calculates the maximum directivity and total radiated power.



模块 9.5 双偶极子阵列 给定两个垂直偶极子，用户可以指定它们各自的长度和电流最大值，以及它们之间的距离和电流激励之间的相位差。该模块生成远区的场和功率图，并计算最大方向性和总辐射功率。



$\psi_1 = -\pi/2$, Eq. (9.104) becomes

$$F_a(\theta) = \left|\sum_{i=0}^{1} a_i e^{j\psi_i} e^{jikd\cos\theta}\right|^2 = \left|1 + e^{-j\pi/2} e^{j(2\pi/\lambda)(\lambda/2)\cos\theta}\right|^2$$
$$= \left|1 + e^{j(\pi\cos\theta - \pi/2)}\right|^2.$$

A function of the form $|1 + e^{jx}|^2$ can be evaluated by factoring out $e^{jx/2}$ from both terms:

$$|1 + e^{jx}|^2 = |e^{jx/2}(e^{-jx/2} + e^{jx/2})|^2$$
$$= |e^{jx/2}|^2 \, |e^{-jx/2} + e^{jx/2}|^2$$
$$= |e^{jx/2}|^2 \left|2\,\frac{[e^{-jx/2} + e^{jx/2}]}{2}\right|^2.$$

The absolute value of $e^{jx/2}$ is 1, and we recognize the function inside the square bracket as $\cos(x/2)$. Hence,

$$|1 + e^{jx}|^2 = 4\cos^2\left(\frac{x}{2}\right).$$

Applying this result to the expression for $F_a(\theta)$, we have

$$F_a(\theta) = 4\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

The power density radiated by the array is then

$$S(R,\theta) = S_0 F_a(\theta) = 4S_0\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

This function has a maximum value $S_{max} = 4S_0$, and it occurs when the argument of the cosine function is equal to zero. Thus,

$$\frac{\pi}{2}\cos\theta - \frac{\pi}{4} = 0,$$

which leads to the solution $\theta = 60°$. Upon normalizing $S(R,\theta)$ by its maximum value, we obtain the normalized radiation intensity given by

$$F(\theta) = \frac{S(R,\theta)}{S_{max}} = \cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

The pattern of $F(\theta)$ is shown in **Fig. 9-28(c)**.

Module 9.6 Detailed Analysis of Two-Dipole Array This module extends the display and computational capabilities of Module 9.5 by offering plots for individual components of **E** and **H** at any range from the antenna, including the near field.

Example 9-6: Pattern Synthesis

In Example 9-5, we were given the array parameters a_0, a_1, ψ_0, ψ_1, and d, and we were then asked to determine the pattern of the two-element dipole array. We now consider the reverse process; given specifications on the desired pattern, we specify the array parameters to meet those specifications.

Given two vertical dipoles, as depicted in **Fig. 9-28(b)**, specify the array parameters so that the array exhibits maximum radiation toward the east and no radiation toward the north or south.

Solution: From Example 9-5, we established that, because each dipole radiates equally along all directions in the y–z plane, the radiation pattern of the two-dipole array in that plane is governed solely by the array factor $F_a(\theta)$. The shape of the pattern of the array factor depends on three parameters: the amplitude ratio a_1/a_0, the phase difference $\psi_1 - \psi_0$, and the spacing d (**Fig. 9-29(a)**). For convenience, we choose $a_0 = 1$ and $\psi_0 = 0$. Accordingly, Eq. (9.101) becomes

$$F_a(\theta) = \left|\sum_{i=0}^{1} a_i e^{j\psi_i} e^{jikd\cos\theta}\right|^2 = |1 + a_1 e^{j\psi_1} e^{j(2\pi d/\lambda)\cos\theta}|^2.$$

Next, we consider the specification that F_a be equal to zero when $\theta = 90°$ (north and south directions in **Fig. 9-29(a)**). For any observation point on the y axis, the ranges R_0 and R_1 shown in **Fig. 9-29(a)** are equal, which means that the propagation phases associated with the time travel of the waves radiated by the two dipoles to that point are identical. Hence, to satisfy the stated condition, we need to choose $a_1 = a_0$ and $\psi_1 = \pm\pi$. With these choices, the signals radiated by the two dipoles have equal amplitudes and opposite phases—thereby interfering destructively. This conclusion can be ascertained by evaluating the array factor at $\theta = 90°$ with $a_0 = a_1 = 1$ and $\psi_1 = \pm\pi$:

$$F_a(\theta = 90°) = |1 + 1e^{\pm j\pi}|^2 = |1 - 1| = 0.$$

The two values of ψ_1, namely π and $-\pi$, lead to the same solution for the value of the spacing d to meet the specification that the array radiation pattern is maximum toward the east, corresponding to $\theta = 0°$. Let us choose $\psi_1 = -\pi$ and examine the array factor at $\theta = 0°$:

$$F_a(\theta = 0) = |1 + e^{-j\pi} e^{j2\pi d/\lambda}|^2 = |1 + e^{j(-\pi + 2\pi d/\lambda)}|^2.$$

For $F_a(\theta = 0)$ to be a maximum, we require the phase angle of the second term to be zero or a multiple of 2π. That is,

$$-\pi + \frac{2\pi d}{\lambda} = 2n\pi,$$

$\Psi 1 = -\pi/2$，等式。 (9.104) 变为

$$F_a(\theta) = \left|\sum_{i=0}^{1} a_i e^{j\psi_i} e^{jikd\cos\theta}\right|^2 = \left|1 + e^{-j\pi/2} e^{j(2\pi/\lambda)(\lambda/2)\cos\theta}\right|^2$$
$$= \left|1 + e^{j(\pi\cos\theta - \pi/2)}\right|^2.$$

$|1 + e^{jx}|^2$ 形式的函数可以通过从两项中分解出 $e^{jx/2}$ 来计算：

$$|1 + e^{jx}|^2 = |e^{jx/2}(e^{-jx/2} + e^{jx/2})|^2$$
$$= |e^{jx/2}|^2 \, |e^{-jx/2} + e^{jx/2}|^2$$
$$= |e^{jx/2}|^2 \left|2\,\frac{[e^{-jx/2} + e^{jx/2}]}{2}\right|^2.$$

$e^{jx/2}$ 的绝对值为 1，我们将方括号内的函数识别为 cos(x/2)。因此，

$$|1 + e^{jx}|^2 = 4\cos^2\left(\frac{x}{2}\right).$$

将此结果应用于 $F_a(\theta)$ 的表达式，我们有

$$F_a(\theta) = 4\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

则阵列辐射的功率密度为

$$S(R,\theta) = S_0 F_a(\theta) = 4S_0\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

该函数具有最大值 $S_{max} = 4S_0$，并且当余弦函数的参数等于 0 时出现。因此，

$$\frac{\pi}{2}\cos\theta - \frac{\pi}{4} = 0,$$

这导致解 $\theta = 60°$。通过最大值对 S(R, θ) 进行归一化，我们得到归一化辐射强度：

$$F(\theta) = \frac{S(R,\theta)}{S_{max}} = \cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right).$$

F(θ)的模式如图9-28(c)所示。

模块 9.6 二偶极子阵列的详细分析 该模块通过提供距天　线任意范围（包括近场）的 E 和 H 各个分量的绘图，扩展了模块 9.5 的显示和计算能力。

例 9-6：模式合成

在示例 9-5 中，我们获得了数组参数 a_0、a_1、ψ_0、ψ_1 和 d，然后要求我们确定二元偶极子数组的模式。我们现在考虑相反的过程；给定所需模式的规格后，我们指定数组参数以满足这些规格。

给定两个垂直偶极子，如图 9-28(b) 所示，指定阵列参数，使阵列向东呈现最大辐射，而向北或向南没有辐射。

解决方案：从示例 9-5 中，我们确定，由于每个偶极子沿 y-z 平面中的所有方向均匀辐射，因此该平面中双偶极子阵列的辐射方向图仅由阵列因子 $F_a(\theta)$ 控制。阵列因子的图案形状取决于三个参数：幅度比a_1/a_0、相位差$\psi_1 - \psi_0$和间距d（图9-29（a））。为了方便起见，我们选择 $a_0 = 1$ 和 $\psi_0 = 0$。 (9.101) 变为

$$F_a(\theta) = \left|\sum_{i=0}^{1} a_i e^{j\psi_i} e^{jikd\cos\theta}\right|^2 = |1 + a_1 e^{j\psi_1} e^{j(2\pi d/\lambda)\cos\theta}|^2.$$

接下来，我们考虑图 9-29(a)) 中 $\theta = 90°$（南北方向时 F_a 等于 0 的规定。对于 y 轴上的任意观测点，图 9-29（a）所示的范围 R_0 和 R_1 相等，这意味着两个偶极子辐射到该点的波的时间传播相关的传播相位是相同的。因此，为了满足规定的条件，我们需要选择 $a_1 = a_0$ 和 $\psi_1 = \pm\pi$。通过这些选择，两个偶极子辐射的信号具有相等的幅度和相反的相位，从而产生破坏性干扰。这个结论可以通过使用 $a_0 = a_1 = 1$ 和 $\psi_1 = \pm\pi$ 评估 $\theta = 90°$ 处的数组因子来确定：

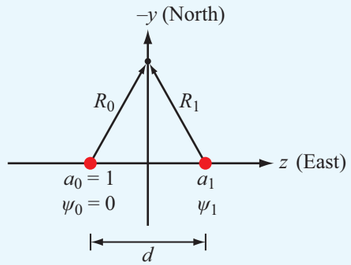
$$F_a(\theta = 90°) = |1 + 1e^{\pm j\pi}|^2 = |1 - 1| = 0.$$

ψ_1 的两个值，即 π 和 $-\pi$，对于间距 d 的值得出相同的解，以满足阵列辐射方向图向东最大的规定，对应于 $\theta = 0°$。让我们选择 $\psi_1 = -\pi$ 并检查 $\theta = 0°$ 处的数组因子：

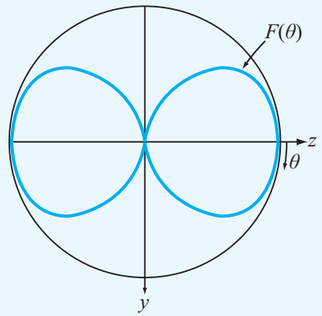
$$F_a(\theta = 0) = |1 + e^{-j\pi} e^{j2\pi d/\lambda}|^2 = |1 + e^{j(-\pi + 2\pi d/\lambda)}|^2.$$

为了使 $F_a(\theta = 0)$ 成为最大值，我们要求第二项的相位角为零或 2π 的倍数。那是，

$$-\pi + \frac{2\pi d}{\lambda} = 2n\pi,$$



(a) Array arrangement



(b) Array pattern

Figure 9-29 (a) Two vertical dipoles separated by a distance d along the z axis; (b) normalized array pattern in the y–z plane for $a_0 = a_1 = 1$, $\psi_1 = \psi_0 = -\pi$, and $d = \lambda/2$.

or

$$d = (2n+1)\frac{\lambda}{2}, \qquad n = 0, 1, 2, \ldots$$

In summary, the two-dipole array will meet the given specifications if $a_0 = a_1$, $\psi_1 - \psi_0 = -\pi$, and $d = (2n+1)\lambda/2$.

For $d = \lambda/2$, the array factor is

$$\begin{aligned} F_a(\theta) &= |1 + e^{-j\pi} e^{j\pi\cos\theta}|^2 \\ &= |1 - e^{j\pi\cos\theta}|^2 \\ &= \left| 2je^{-j(\pi/2)\cos\theta} \left[\frac{e^{j(\pi/2)\cos\theta} - e^{-j(\pi/2)\cos\theta}}{2j} \right] \right|^2 \\ &= 4\sin^2\left(\frac{\pi}{2}\cos\theta\right). \end{aligned}$$

The array factor has a maximum value of 4, which is the maximum level attainable from a two-element array with unit amplitudes. The directions along which $F_a(\theta)$ is a maximum are those corresponding to $\theta = 0$ (east) and $\theta = 180°$ (west), as shown in **Fig. 9-29(b)**.

Exercise 9-14: Derive an expression for the array factor of a two-element array excited in phase with $a_0 = 1$ and $a_1 = 3$. The elements are positioned along the z axis and are separated by $\lambda/2$.

Answer: $F_a(\theta) = [10 + 6\cos(\pi\cos\theta)]$. (See EM.)

Exercise 9-15: An equally spaced N-element array arranged along the z axis is fed with equal amplitudes and phases; that is, $A_i = 1$ for $i = 0, 1, \ldots, (N-1)$. What is the magnitude of the array factor in the broadside direction?

Answer: $F_a(\theta = 90°) = N^2$. (See EM.)

9-10 *N*-Element Array with Uniform Phase Distribution

We now consider an array of N elements with equal spacing d and equal-phase excitations; that is, $\psi_i = \psi_0$ for $i = 1, 2, \ldots, (N-1)$. Such an array of in-phase elements is sometimes referred to as a ***broadside array*** because the main beam of the radiation pattern of its array factor is always in the direction broadside to the array axis. From Eq. (9.104), the array factor is given by

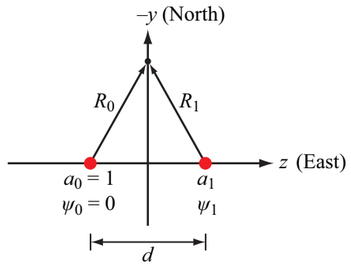
$$\begin{aligned} F_a(\theta) &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2 \\ &= |e^{j\psi_0}|^2 \left| \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2. \end{aligned} \tag{9.105}$$

The phase difference between the fields radiated by adjacent elements is

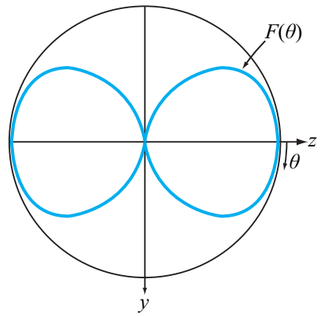
$$\gamma = kd\cos\theta = \frac{2\pi d}{\lambda}\cos\theta. \tag{9.106}$$

In terms of γ, Eq. (9.105) takes the compact form

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2. \quad \textbf{(uniform phase)} \tag{9.107}$$



(a) Array arrangement



(b) Array pattern

图9-29(a)沿z轴间隔距离d的两个垂直偶极子；(b) $a_0 = a_1 = 1$、$\psi_1 = \psi_0 = -\pi$ 和 $d = \lambda/2$ 在 y-z 平面中的归一化阵列模式。

或者

$$d = (2n+1)\frac{\lambda}{2}, \qquad n = 0, 1, 2, \ldots$$

总之，如果 $a_0 = a_1$、$\psi_1 - \psi_0 = -\pi$ 和 $d = (2n+1)\lambda/2$，则二偶极子阵列将满足给定的规格。

对于 $d = \lambda/2$，数组因子为

$$\begin{aligned} F_a(\theta) &= |1 + e^{-j\pi} e^{j\pi\cos\theta}|^2 \\ &= |1 - e^{j\pi\cos\theta}|^2 \\ &= \left| 2je^{-j(\pi/2)\cos\theta} \left[\frac{e^{j(\pi/2)\cos\theta} - e^{-j(\pi/2)\cos\theta}}{2j} \right] \right|^2 \\ &= 4\sin^2\left(\frac{\pi}{2}\cos\theta\right). \end{aligned}$$

阵列因子的最大值为 4，这是具有单位幅度的二元阵列可达到的最大水平。$F_a(\theta)$最大的方向是$\theta = 0$（东）和$\theta = 180°$ (西)对应的方向，如图9-29（b）所示。

练习 9-14：推导出与 $a_0 = 1$ 和 $a_1 = 3$ 同相激励的二元阵列的阵列因子表达式。元素沿 z 轴定位，并以 $\lambda/2$ 分隔。

Answer: $F_a(\theta) = [10 + 6\cos(\pi\cos\theta)]$. (See EM.)

练习 9-15：沿 z 轴排列的等距 N 元阵列具有相等的幅度和相位；即，$A_i = 1$ 对于 $i = 0, 1, \ldots, (N-1)$。侧面方向上的阵列因子的大小是多少？

Answer: $F_a(\theta = 90°) = N^2$. (See EM.)

9-10 具有均匀相位分布的 N 元阵列

现在我们考虑一个由 N 个元素组成的阵列，其具有等间距 d 和等相位激励；即，$\psi_i = \psi_0$ 代表 $i = 1, 2, \ldots, (N-1)$。这种同相元件的阵列有时被称为侧边阵列，因为其阵列因子的辐射方向图的主波束总是在阵列轴的侧边方向上。从方程。(9.104)，阵列因子由下式给出

$$\begin{aligned} F_a(\theta) &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2 \\ &= |e^{j\psi_0}|^2 \left| \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{jikd\cos\theta} \right|^2. \end{aligned} \tag{9.105}$$

相邻元件辐射的场之间的相位差为

$$\gamma = kd\cos\theta = \frac{2\pi d}{\lambda}\cos\theta. \tag{9.106}$$

就 γ 而言，等式：(9.105) 采取紧凑形式

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2. \quad \textbf{(uniform phase)} \tag{9.107}$$

For a uniform amplitude distribution with $a_i = 1$ for $i = 0, 1, \ldots, (N-1)$, Eq. (9.107) becomes

$$F_a(\gamma) = |1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma}|^2. \tag{9.108}$$

This geometric series can be rewritten in a more compact form by applying the following recipe. First, we define

$$F_a(\gamma) = |f_a(\gamma)|^2, \tag{9.109}$$

with

$$f_a(\gamma) = [1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma}]. \tag{9.110}$$

Next, we multiply $f_a(\gamma)$ by $e^{j\gamma}$ to obtain

$$f_a(\gamma)\, e^{j\gamma} = (e^{j\gamma} + e^{j2\gamma} + \cdots + e^{jN\gamma}). \tag{9.111}$$

Subtracting Eq. (9.111) from Eq. (9.110) gives

$$f_a(\gamma)\,(1 - e^{j\gamma}) = 1 - e^{jN\gamma}, \tag{9.112}$$

which in turn gives

$$f_a(\gamma) = \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \frac{(e^{-jN\gamma/2} - e^{jN\gamma/2})}{(e^{-j\gamma/2} - e^{j\gamma/2})}$$
$$= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \tag{9.113}$$

After multiplying $f_a(\gamma)$ by its complex conjugate, we obtain the result:

$$F_a(\gamma) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \tag{9.114}$$

(uniform amplitude and phase)

From Eq. (9.108), $F_a(\gamma)$ is maximum when all terms are 1, which occurs when $\gamma = 0$ (or equivalently, $\theta = \pi/2$). Moreover, $F_a(0) = N^2$. Hence, the normalized array factor is given by

$$F_{an}(\gamma) = \frac{F_a(\gamma)}{F_{a,\max}}$$
$$= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)} = \frac{\sin^2\left(\frac{N\pi d}{\lambda}\cos\theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda}\cos\theta\right)}. \tag{9.115}$$

A polar plot of $F_{an}(\theta)$ is shown in **Fig. 9-30** for $N = 6$ and $d = \lambda/2$. The reader is reminded that this is a plot of the radiation pattern of the array factor alone; the pattern for the antenna array is equal to the product of this pattern and that of a single element, as discussed earlier in connection with the pattern multiplication principle.

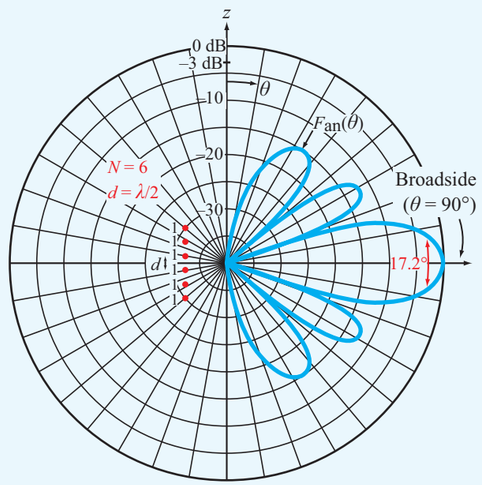


Figure 9-30 Normalized array pattern of a uniformly excited six-element array with interelement spacing $d = \lambda/2$.

Example 9-7: Multiple-Beam Array

Obtain an expression for the array factor of a two-element array with equal excitation and a separation $d = 7\lambda/2$, and then plot the array pattern.

Solution: The array factor of a two-element array ($N = 2$) with equal excitation ($a_0 = a_1 = 1$) is given by

$$F_a(\gamma) = \left|\sum_{i=0}^{1} a_i e^{ji\gamma}\right|^2 = |1 + e^{j\gamma}|^2,$$
$$= |e^{j\gamma/2}(e^{-j\gamma/2} + e^{j\gamma/2})|^2$$
$$= |e^{j\gamma/2}|^2\, |e^{-j\gamma/2} + e^{j\gamma/2}|^2 = 4\cos^2(\gamma/2),$$

where $\gamma = (2\pi d/\lambda)\cos\theta$. The normalized array pattern shown in **Fig. 9-31** consists of seven beams; all have the same peak value but not the same angular width. The number of beams in the angular range between $\theta = 0$ and $\theta = \pi$ is equal to the

对于均匀振幅分布，$a_i = 1$ 对于 $i = 0, 1, \ldots, (N-1)$，方程：(9.107) 变为

$$F_a(\gamma) = |1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma}|^2. \tag{9.108}$$

通过应用以下配方，可以将这个几何级数重写为更紧凑的形式。首先，我们定义

$$F_a(\gamma) = |f_a(\gamma)|^2, \tag{9.109}$$

和

$$f_a(\gamma) = [1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma}]. \tag{9.110}$$

接下来，我们将 $f_a(\gamma)$ 乘以 $e^{j\gamma}$ 得到

$$f_a(\gamma)\, e^{j\gamma} = (e^{j\gamma} + e^{j2\gamma} + \cdots + e^{jN\gamma}). \tag{9.111}$$

减法(9.111) 来自方程式(9.110) 给出

$$f_a(\gamma)\,(1 - e^{j\gamma}) = 1 - e^{jN\gamma}, \tag{9.112}$$

这又给出了

$$f_a(\gamma) = \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \frac{(e^{-jN\gamma/2} - e^{jN\gamma/2})}{(e^{-j\gamma/2} - e^{j\gamma/2})}$$
$$= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \tag{9.113}$$

将 $f_a(\gamma)$ 与其复共轭相乘后，我们得到结果：

$$F_a(\gamma) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \tag{9.114}$$

(均匀的幅度和相位)

从方程。(9.108)，当所有项均为 1 时，$F_a(\gamma)$ 最大，这发生在 $\gamma = 0$（或等效地，$\theta = \pi/2$）时。此外，$F_a(0) = N^2$。因此，归一化数组因子由下式给出

$$F_{an}(\gamma) = \frac{F_a(\gamma)}{F_{a,\max}}$$
$$= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)} = \frac{\sin^2\left(\frac{N\pi d}{\lambda}\cos\theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda}\cos\theta\right)}. \tag{9.115}$$

$N = 6$ 和 $d = \lambda/2$ 的 $F_{an}(\theta)$ 极坐标图如图 9-30 所示。提醒读者，这是一个情节

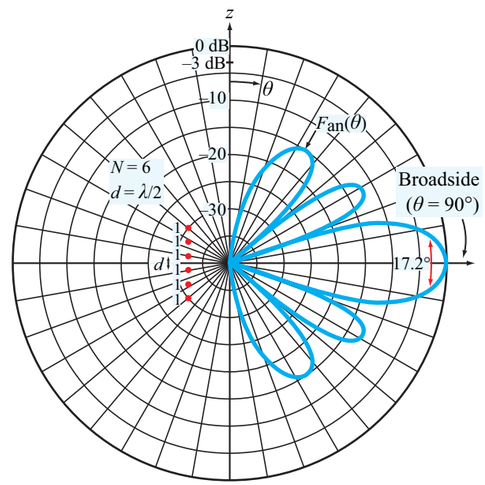


图 9-30 元素间间距为 $d = \lambda/2$ 的均匀激发六元素阵列的归一化阵列图案。

单独的阵列因子的辐射方向图；天线阵列的方向图等于该方向图与单个元件的方向图的乘积，如前面结合方向图乘法原理所讨论的。

示例 9-7：多光束阵列

获取具有相等激励和间隔 $d = 7\lambda/2$ 的二元阵列的阵列因子的表达式，然后绘制阵列模式。

解：具有相等激励 ($a_0 = a_1 = 1$) 的二元阵列 ($N = 2$) 的阵列因子由下式给出

$$F_a(\gamma) = \left|\sum_{i=0}^{1} a_i e^{ji\gamma}\right|^2 = |1 + e^{j\gamma}|^2,$$
$$= |e^{j\gamma/2}(e^{-j\gamma/2} + e^{j\gamma/2})|^2$$
$$= |e^{j\gamma/2}|^2\, |e^{-j\gamma/2} + e^{j\gamma/2}|^2 = 4\cos^2(\gamma/2),$$

其中 $\gamma = (2\pi d/\lambda)\cos\theta$。图9-31所示的归一化阵列图案由七个波束组成；都具有相同的峰值，但角宽度不同。 $\theta = 0$ 和 $\theta = \pi$ 之间的角度范围内的光束数量等于